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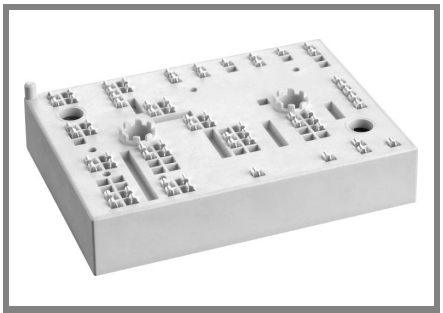
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SKIIP 35NAB12T4V1

EN
This Datasheet is presented by the manufacturer

DE
Dieses Datenblatt wird vom Hersteller bereitgestellt

FR
Cette fiche technique est présentée par le fabricant



MiniSKiiP® 3

3-phase bridge rectifier + brake chopper + 3-phase bridge inverter

SKiiP 35NAB12T4V1

Features

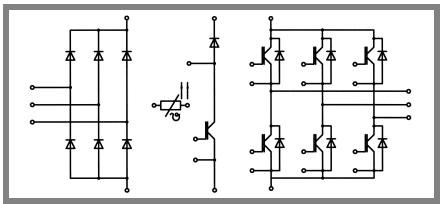
- Trench 4 IGBT´s
- Robust and soft freewheeling diodes in CAL technology
- Highly reliable spring contacts for electrical connections
- UL recognised file no. E63532

Typical Applications*

- Inverter up to 26 kVA
- Typical motor power 15 kW

Remarks

- V_{CEsat} , V_F= chip level value
- Case temp. limited to T_C = 125°C max. (for baseplateless modules $T_C = T_S$)
- product rel. results valid for $T_j \leq 150$ (recomm. T_{op} = -40 ... +150°C)



NAB

Absolute Maximum Ratings	T_s = 25 °C, unless otherwise specified		
Symbol	**Conditions**	**Values**	**Units**
IGBT - Inverter, Chopper			
V_{CES}		1200	V
I_C	T_s = 25 (70) °C	67 (52)	A
I_{CRM}		150	A
V_{GES}		± 20	V
T_j		- 40 ... + 175	°C
Diode - Inverter, Chopper			
I_F	T_s = 25 (70) °C	60 (48)	A
I_{FRM}		150	A
T_j		- 40 ... + 175	°C
Diode - Rectifier			
V_{RRM}		1600	V
I_F	T_s = 70 °C	61	A
I_{FSM}	t_p = 10 ms, sin 180 °, T_j = 25 °C	700	A
i^2t	t_p = 10 ms, sin 180 °, T_j = 25 °C	2400	A^2s
T_j		- 40 ... + 150	°C
Module			
I_{tRMS}	per power terminal (20 A / spring)	80	A
T_{stg}		- 40 ... + 125	°C
V_{isol}	AC, 1 min.	2500	V

Characteristics	T_s = 25 °C, unless otherwise specified				
Symbol	**Conditions**	**min.**	**typ.**	**max.**	**Units**
IGBT - Inverter, Chopper					
V_{CEsat}	I_{Cnom} = 50 A, T_j = 25 (150) °C		1,85 (2,25)	2,05 (2,45)	V
$V_{GE(th)}$	$V_{GE} = V_{CE}$, I_C = 2 mA	5	5,8	6,5	V
$V_{CE(TO)}$	T_j = 25 (150) °C		0,8 (0,7)	0,9 (0,8)	V
r_T	T_j = 25 (150) °C		21 (31)	23 (33)	mΩ
C_{ies}	V_{CE} = 25 V, V_{GE} = 0 V, f = 1 MHz		2,77		nF
C_{oes}	V_{CE} = 25 V, V_{GE} = 0 V, f = 1 MHz		0,21		nF
C_{res}	V_{CE} = 25 V, V_{GE} = 0 V, f = 1 MHz		0,16		nF
$R_{th(j-s)}$	per IGBT		0,71		K/W
$t_{d(on)}$	under following conditions		60		ns
t_r	V_{CC} = 600 V, V_{GE} = ± 15 V		35		ns
$t_{d(off)}$	I_{Cnom} = 50 A, T_j = 150°C		370		ns
t_f	$R_{Gon} = R_{Goff}$ = 15 Ω		60		ns
E_{on}	inductive load		6		mJ
E_{off}			4,7		mJ
Diode - Inverter, Chopper					
$V_F = V_{EC}$	I_{Fnom} = 50 A, T_j = 25 (150) °C		2,25 (2,2)	2,55 (2,5)	V
$V_{(TO)}$	T_j = 25 (150) °C		1,3 (0,9)	1,5 (1,1)	V
r_T	T_j = 25 (150) °C		19 (26)	21 (28)	mΩ
$R_{th(j-s)}$	per diode		0,95		K/W
I_{RRM}	under following conditions		45		A
Q_{rr}	I_{Fnom} = 50 A, V_R = 600 V		8,6		µC
E_{rr}	V_{GE} = 0 V, T_j = 150 °C		3,4		mJ
	di_F/dt = 1400 A/µs				
Diode - Rectifier					
V_F	I_{Fnom} = 35 A, T_j = 25 °C		1,1		V
$V_{(TO)}$	T_j = 150 °C		0,8		V
r_T	T_j = 150 °C		11		mΩ
$R_{th(j-s)}$	per diode		0,9		K/W
Temperature Sensor					
R_{ts}	3 %, T_r = 25 (100) °C		1000(1670)		Ω
Mechanical Data					
w			95		g
M_s	Mounting torque	2		2,5	Nm

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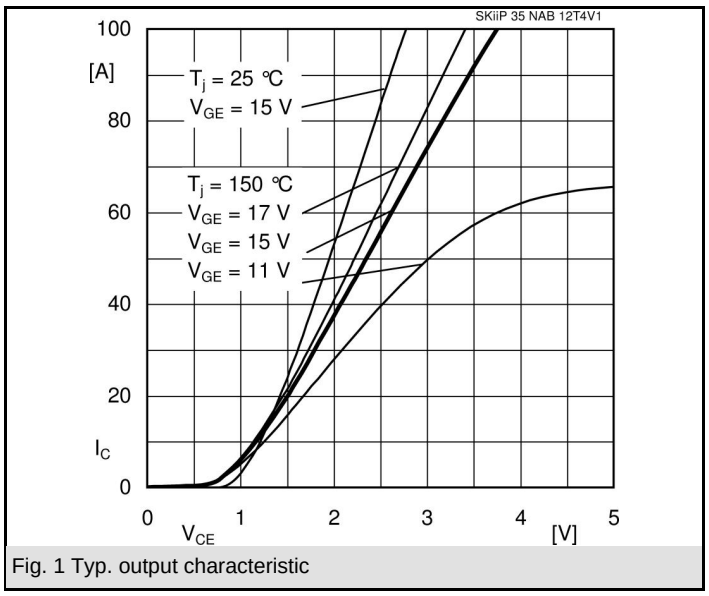


Fig. 1 Typ. output characteristic

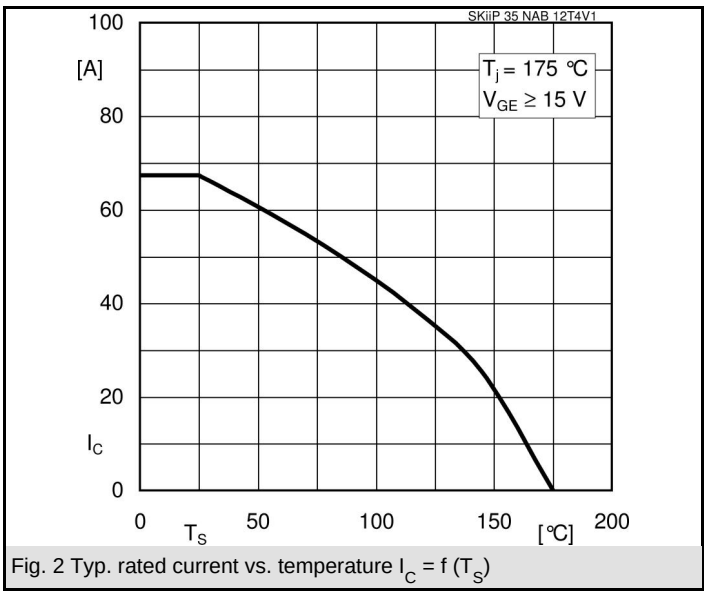


Fig. 2 Typ. rated current vs. temperature $I_C = f(T_S)$

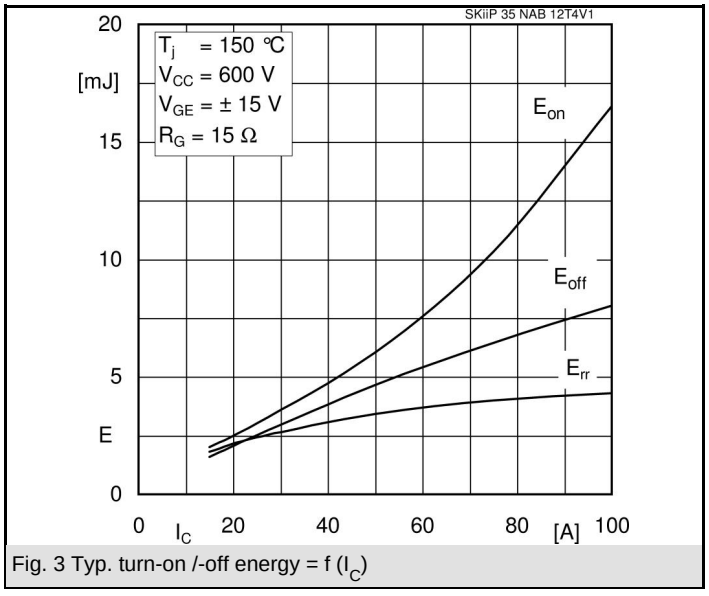


Fig. 3 Typ. turn-on /-off energy = f (I_C)

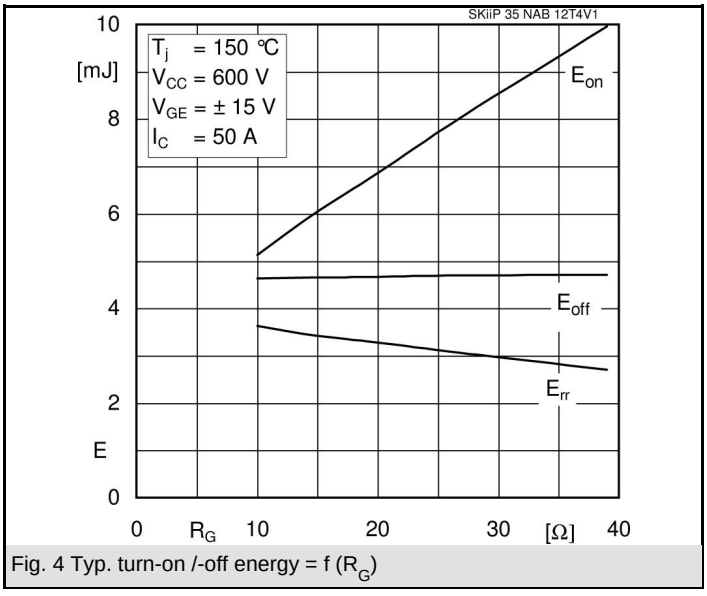


Fig. 4 Typ. turn-on /-off energy = f (R_G)

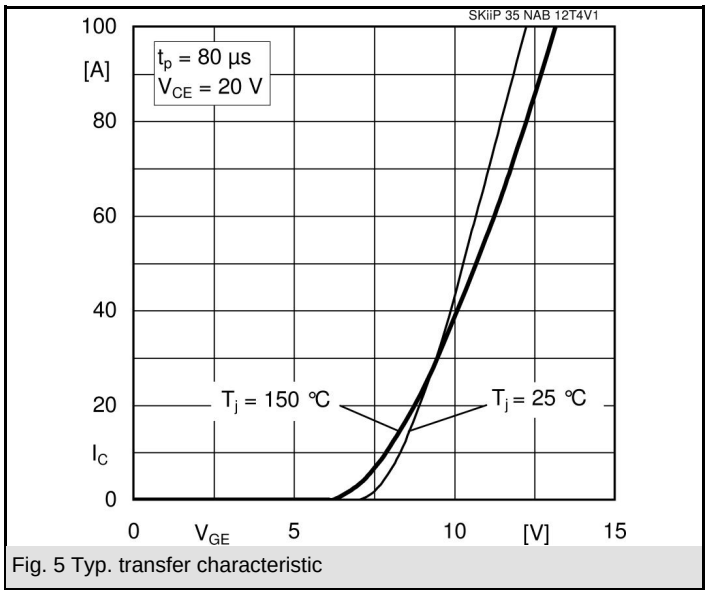


Fig. 5 Typ. transfer characteristic

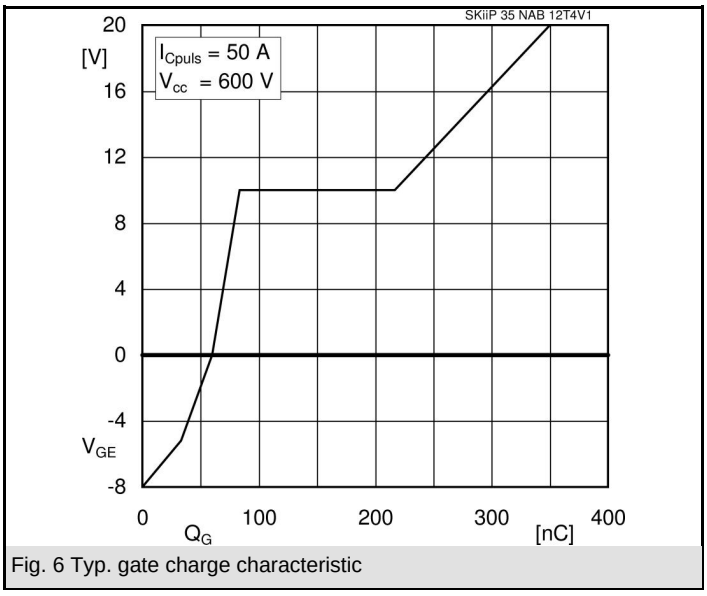


Fig. 6 Typ. gate charge characteristic

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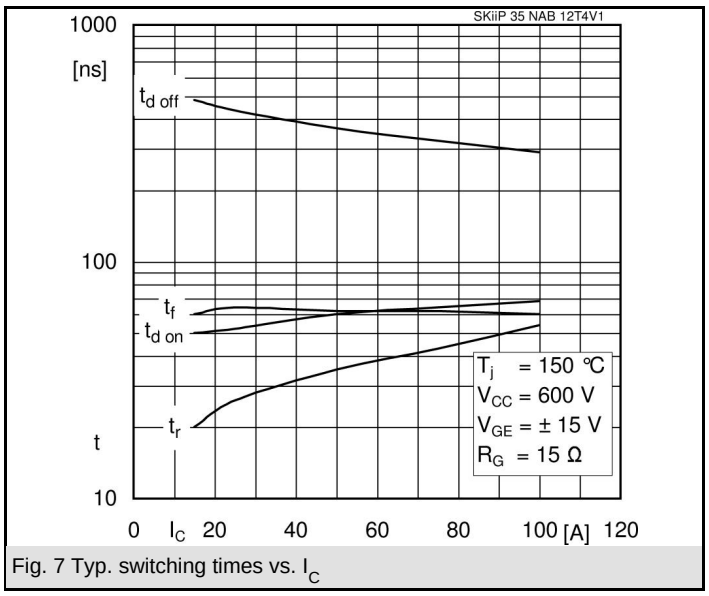


Fig. 7 Typ. switching times vs. I_C

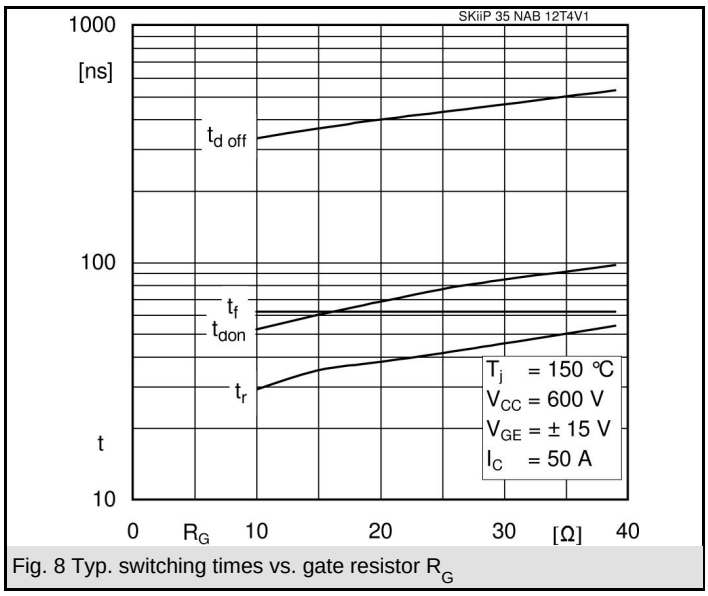


Fig. 8 Typ. switching times vs. gate resistor R_G

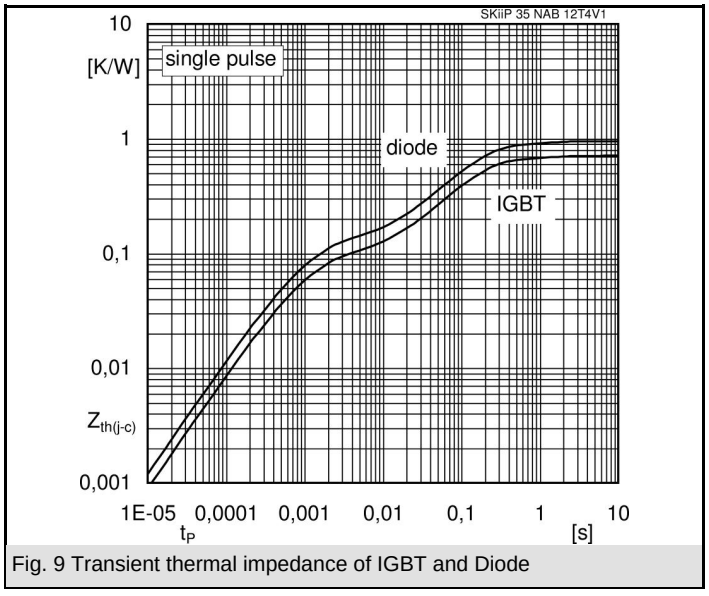


Fig. 9 Transient thermal impedance of IGBT and Diode

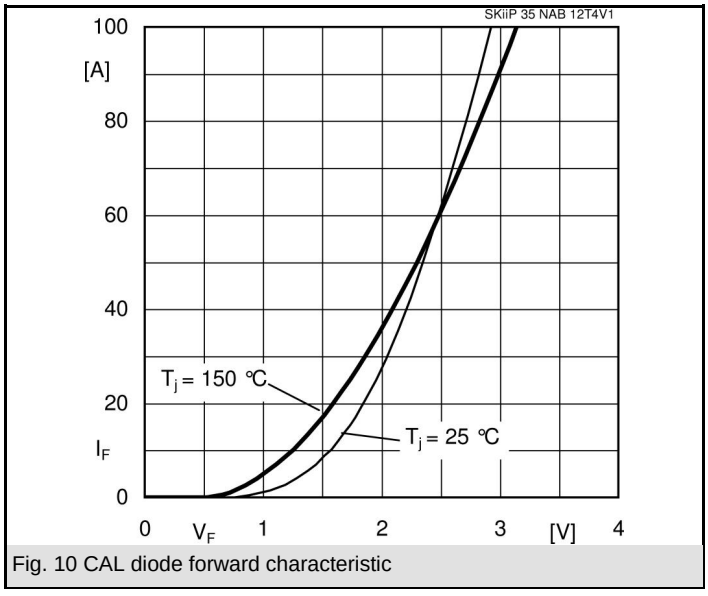


Fig. 10 CAL diode forward characteristic

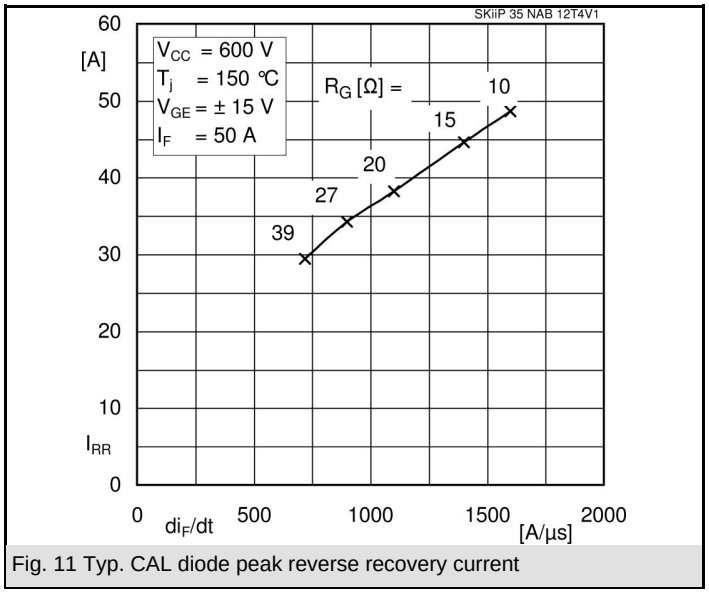


Fig. 11 Typ. CAL diode peak reverse recovery current

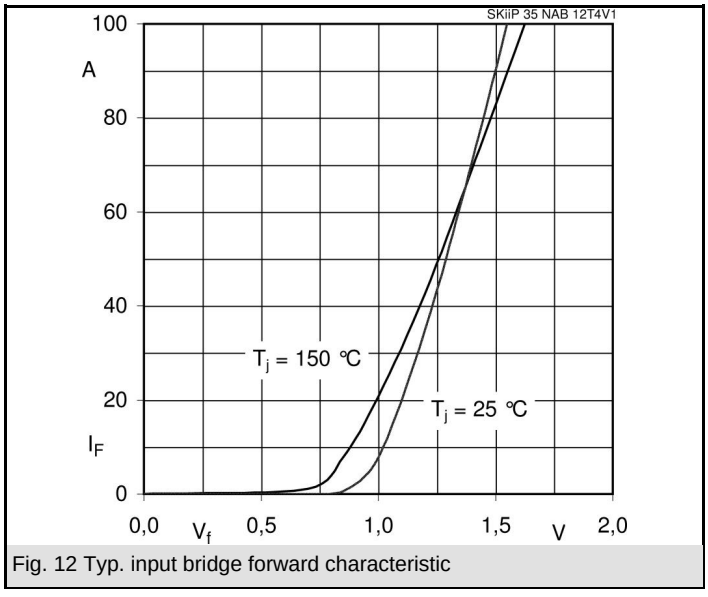
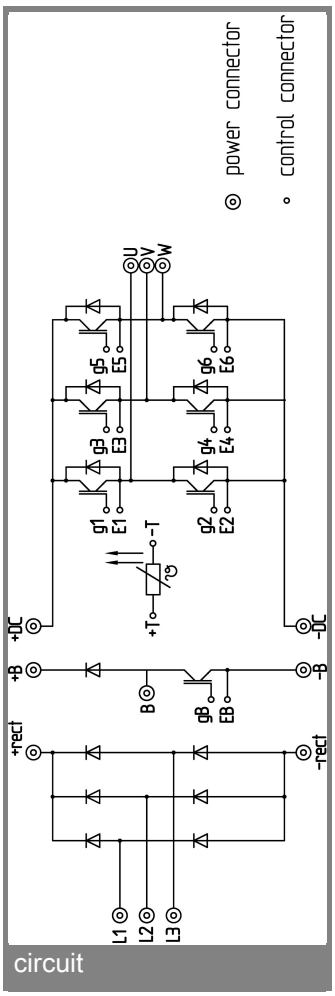
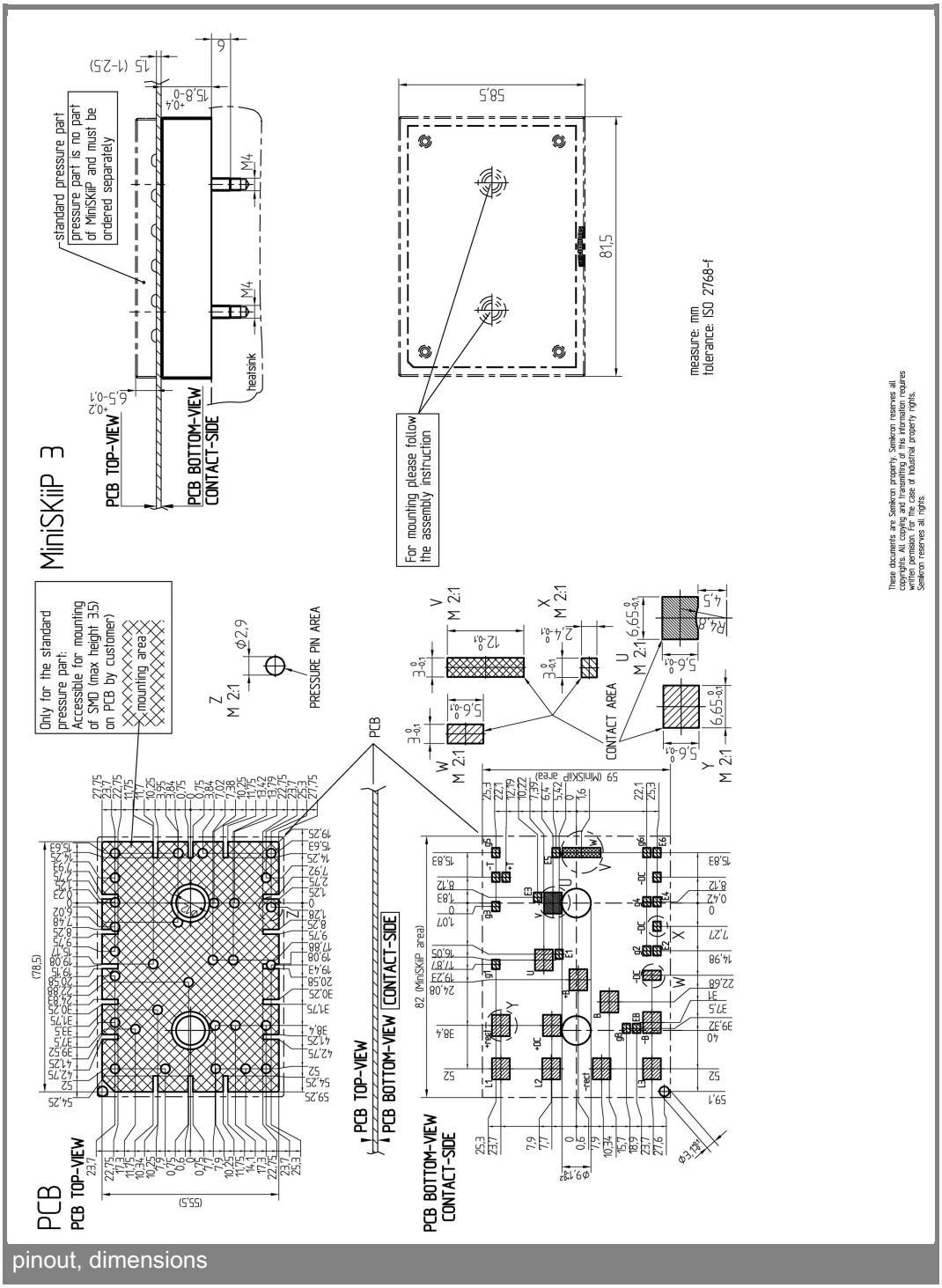


Fig. 12 Typ. input bridge forward characteristic

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circuit



pinout, dimensions

This is an electrostatic discharge sensitive device (ESDS), international standard IEC 60747-1, Chapter IX.

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